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# Applying evolutionary theory to understand host–microbiome evolution

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All plants and animals are host to a community of microorganisms, their microbiotas, which have crucial influences on the life history and performance of their hosts. Despite the importance of such host–microbiota relationships, relatively little is known about the role microbiotas have in mediating evolution of the host and entire host–microbe assemblages. This knowledge gap is partly due to the lack of theoretical frameworks that generate testable predictions on the evolutionary dynamics of host–microbiota systems. In this Perspective, we argue that the foundation for such frameworks exists in evolutionary theory. We highlight four examples of theoretical models—niche construction, indirect genetic effects, maternal effects and multilevel selection—that capture important aspects of host–microbiome evolution. We outline how each of these frameworks can provide key insights into the evolution of host–microbiota systems while also suggesting expansions of current theory to incorporate processes unique to host–microbe assemblages; for instance, focusing on nuances in microbiota transmission and ecological microbial community dynamics. Expanding evolutionary theory to accommodate host–microbiota associations is key for a more integrative understanding of evolution, and the approach outlined here can guide future empirical research on the function and evolution of these ubiquitous interactions.

All multicellular organisms have long-standing, intimate relationships with microorganisms. These host-associated microbial communities (including bacteria, archaea, viruses, protists and fungi—together termed the microbiota) are crucial for host performance, affecting host traits related to metabolism[1], pathogen resistance[2], immune development[3], disease[4] and behaviour[5], among many others. Beyond its fundamental interest, the potential applications of the microbiota vary widely, ranging from human health[6] to sustainable agriculture[7], conservation biology[8] and adaptation to climate change[9].

There is no doubt that host-associated microbiotas can influence host performance. There is also increasing evidence that the association of host and microbiota can facilitate adaptation to new environmental conditions, either through changes in the microbiota, the host or both[10–13]. There is probably variation among host systems in the exact processes mediating such host–microbiome evolution, including host–microbe coevolutionary interactions, as proposed by the hologenome theory of evolution[14], or less strict coevolutionary interactions[12,15–22]. Fidelity of microbiota transmission (Box 1), the multiple levels of selection and evolutionary interests possible in host–microbe associations[22] are all key determinants shaping the evolutionary dynamics of the association.

An in-depth appreciation of the exact characteristics of host–microbiota systems and their evolution greatly benefits from theoretical approaches[23]. Although previous theoretical work has established

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## BOX 1

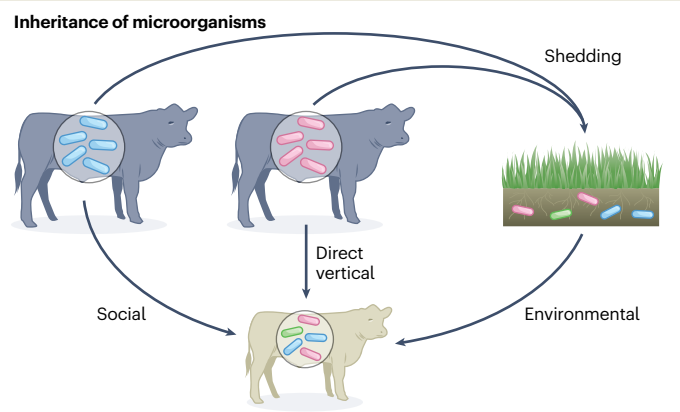
# Mechanisms of microbiome inheritance

The fidelity of the microbiota across host generations is the most critical factor that determines whether microorganisms share the same evolutionary interests as their hosts, thus generally favouring beneficial interactions. Different mechanisms favour fidelity or infidelity in microbiota transmission across host generations (see the figure below and figure 2 in ref. 52).

One process that results in cross-generational host–microbiota fidelity is the vertical transmission of microorganisms from parents to offspring. Strict vertical transmission, akin to genetic inheritance, occurs through intracellular infection of germ cells as observed, for example, in aphid–*Buchnera*[102] or carpenter ant–*Blochmannia* interactions[103] and in more complex associations between sponges and part of their microbiota[104]. However, even in the absence of such strict vertical transmission, 'intimate neighbourhood transmission'[105] may result in the transmission from parents to offspring; for instance, through the covering of eggs with microorganisms[106], through mode of delivery in humans[107], or through hosts shaping microbial community composition of their environment as a form of niche construction[108]. Further, vertical microbiota transmission goes beyond direct transmission from parents to offspring: living in proximity (for example, sharing the same household with relatives) can produce a so-called social microbiota that ultimately promotes microbiota fidelity[54]. However, vertical transmission can be incomplete in complex associations, such as between sponges and their microbiota[104]. Even in the absence of vertical transmission, host genotypes might directly influence the types of microorganism that can be taken up and establish in a particular host, shaping microbiota composition and increasing cross-generational fidelity[109].

Environmental transmission can also result in host–microbe fidelity across generations, whenever hosts faithfully acquire the same microorganisms from the environment every generation, as found in the bobtail squid–*Vibrio*[110] and stinkbug–*Burkholderia* associations[111]. Whenever the environmental microbial pool responds to selection on hosts, environmental acquisition alone can lead to cross-generational microbiota fidelity, through 'collective inheritance'[26]. A combination of these different processes can shape microbiota fidelity in single host systems, as shown in wood mice (vertical and social transmission)[55] and sponges (direct and indirect vertical transmission)[104].

Despite all these different biological processes that may bolster microbiota fidelity, many host-associated microorganisms were proposed to lack cross-generational fidelity[22], and the exact degree of microbiota fidelity is often unknown for most host species. In other cases, certain interventions may compromise vertical microorganism transmission. Examples include antibiotic therapy on newborns[112,113] or Caesarean section (that is, delivery mode), which seems to have a transient effect on microbiota composition in newborns and is then further shaped by breastfeeding (or its absence) and early life exposure to microorganisms[114–116]. To quantify the fidelity of microbiota transmission, one can estimate the heritability of microbiota composition, which is the proportion of microbiota variance (for example, variance in relative abundance of a microbial taxon across hosts in a population) attributable to host genotypic variance. Microbiota heritability has been estimated for only a limited number of plant and animal host species[117], suggesting low microbiota heritabilities in general, although some were on par with heritabilities of important host traits.




**Three primary modes of microbial inheritance.** The left mode indicates cross-generational transmission of microorganisms as a consequence of social acquisition of microorganisms. The middle mode indicates cross-generational transmission of microorganisms due to direct parent–offspring transmission. The right mode indicates microbial inheritance mediated by the environment, where host offspring acquire environmental microorganisms that descend from microorganisms shed by hosts in a previous generation. Additional transmission routes via the environment can also ensure fidelity, as previously described.

new frameworks to study host–microbe systems[24–27], the full breadth of well-established evolutionary theory has not been applied to understand the evolution of host–microbe associations.

We propose taking advantage of existing theory in evolutionary biology to explore and dissect the evolution of host–microbiota systems. We highlight four existing frameworks that address key characteristics of host–microbiome evolutionary dynamics. We discuss how we may borrow useful elements from each of these frameworks, while also highlighting fundamental differences between host–microbe evolutionary dynamics and existing frameworks, pinpointing features of host–microbiome evolution that require the development of new theory. We point to important directions for future theoretical work, while emphasizing the importance of integrating theory and empirical work.

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## Adapting evolutionary frameworks for host–microbiota systems

We define host–microbiome evolution as changes in allele frequencies in the host population and/or in associated microbial communities (through shifts in allele frequencies within microbial lineages or in the relative abundances of microbial taxa) occurring within or across host generations. When host trait variation is explained by genetic variation across either hosts or microbiotas, then host–microbiome evolution can result in changes of host traits over time. A few important processes contributing to host–microbiome evolution include microbial meta-community dynamics[28], microbe–microbe coevolution and host–microbe coevolution[29], and host dispersal. In general, we lack an understanding of the consequences of natural selection on host–microbiota systems and the inheritance of selected host trait variation that is mediated by the inheritance of host-associated microbiotas. In the next sections, we explore how four existing evolutionary frameworks may enhance our understanding of host–microbiome evolution over the timescale in which host microevolutionary dynamics occur (for example, from a single to possibly thousands of host generations), each addressing different aspects of the evolution of host–microbiota systems, and where the choice of framework will depend on the study question. These frameworks are as follows: (1) niche construction, (2) indirect genetic effects (IGEs), (3) maternal effects (MEs), and (4) multilevel selection (MLS).

Each framework considers how microbiotas interact with the evolution of host traits, and treats host-associated microbiotas essentially as a form of non-genetic inheritance (NGI). NGI involves the transmission of factors other than the DNA (for example, epigenetic patterns, cytoplasmic transmission, nutrient provisioning and cultural inheritance), from parents (or other conspecifics) to offspring. Depending on how these non-genetic factors covary and interact with genetic, environmental and/or stochastic factors, NGI can manifest itself as, for instance, MEs[30], ecological inheritance[31], IGEs[32] or possibly some combination of these effects. There exists a large body of literature on the implications of NGI for plant and animal evolution[33]. The fourth framework, MLS describes how composites of individuals respond to selection that jointly acts on various scales of biological organization, and also considers the evolution of microbial traits.

To illustrate concepts, we consider host traits that are jointly mediated by host genomes and host-associated microbiotas. More precisely, we build on previous quantitative genetic approaches by assuming host traits can be additively decomposed into genetic and microbial components[34,35]. Here, microbiotas are summarized by their additive effect on a host trait, measured by a linear model that explains host trait variation in terms of microbiota composition[35–39]. This is analogous to classical quantitative genetic methods for summarizing genomes by their additive genetic effects on traits[40].

Our focal frameworks are not mutually exclusive. In addition, we introduce the concepts with mathematical terms to organize the main underlying principles, to enhance clarity of key relationships or dependencies within the frameworks, and to stimulate further theoretical work. Our symbolic representation can and should be extended to account for additional factors contributing to host–microbiome evolution.

## Niche construction

The framework of niche construction is centred on the reciprocal dynamics of populations and their environment. Niche construction considers the indirect effects of organismal activity on their own or descendants' fitness through environmental modification, and it has been considered an evolutionary process in its own right[41]. There are two interpretations of niche construction, both applicable to host–microbiota systems (Fig. 1): (1) environmental modification by organismal activity (which may or may not have evolutionary consequences for either the host or associated microorganisms), and (2) an evolutionary process involving feedback between environmental change and organismal evolution.

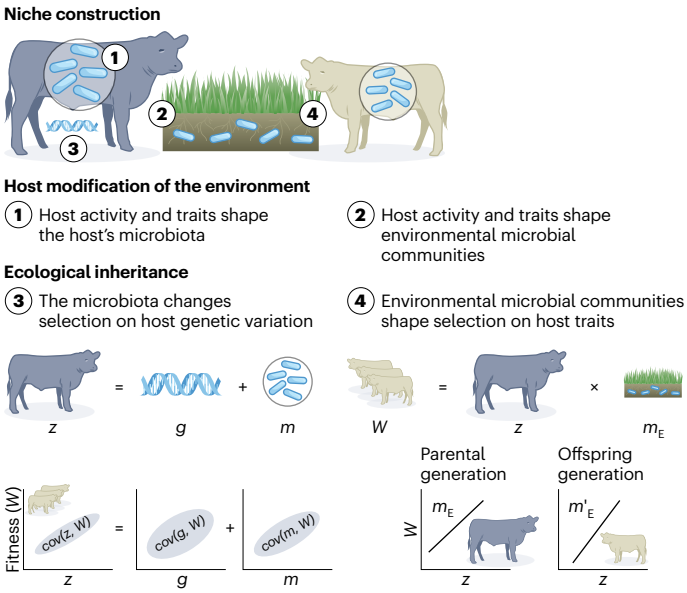



**Fig. 1 | Conceptual overview for applying niche construction to host–microbiome evolution.** Numbers (1) and (2) focus on niche construction simply as environmental modification, and numbers (3) and (4) consider niche construction as an evolutionary process with ecological inheritance. (1) Interpretation of a host's microbiota as a component of the host environment modified by host activity (such as diet and immune response). (2) Interpretation of the microbiota in the host's immediate surroundings as the host environment is modified by activity such as shedding and priming. (3) Selection on microbiota-mediated host traits (quantified by the relationship between fitness $W$ and phenotype $z$, as illustrated at the bottom of number (3)) decomposes into selection on genetic $g$ and microbial $m$ components. Even when selection consistently favours certain values of the host trait $z$, the resulting selection on host genetic variation depends critically on how microbiota variation is structured. (4) Microbiotas of the host's immediate environment can shape selection on host traits (for example, by competing with soil pathogens or altering nutrient availability). Here the slope of host fitness in relation to host trait is determined by the environmental microbiota $m_E$. In this example, host offspring encounter an environment with a steeper $m'_E$ and consequently experience stronger directional selection.

### Niche construction as an environmental modification

Niche construction as environmental modification by organismal activity applies to host–microbiota systems in at least two ways. First, the microbiota of a host (such as microbiotas associated with the host's skin or gut) can be considered a part of the host's environment (Fig. 1, number (1)). Host activity and host traits that result in microorganism acquisition (for example, through feeding, social behaviour or morphology), and host immune responses that result in selection of microorganisms, provide mechanisms of niche construction.

Second, the microbiota of a host's immediate surroundings (for example, microbial communities associated with different food sources or with surfaces the host comes into contact with) can be considered a part of the host's environment (Fig. 1, number (2)). Niche construction then occurs when host activity alters the environmental microbiota composition; for instance, by shedding microorganisms into the surroundings, by host-mediated structuring of the environment (for example, nest building)[42] or the construction of other types of built environment[43], by nutrient provisioning (for example, 'priming' of soil microorganisms by plant roots)[44] and by 'farming' activities (for example, the cultivation of fungi by insects)[45]. Resulting altered

microbiota composition or microbial activity may consequently affect host fitness, for example, by increasing nutrient availability or suppressing pathogens, as has been observed in soil surrounding plant roots[46,47]. Furthermore, if host genotypes show differential fitness in response to such environmental modifications, then niche construction can establish an evolutionary process, as discussed next.

### Niche construction as an evolutionary process

The interpretation of niche construction as an evolutionary process is more stringent as it requires the maintenance of natural selection pressures across generations (Fig. 1, numbers (3) and (4)). This process, called ecological inheritance, requires that the activity of a particular organism modifies the environment, and that these environmental modifications influence selection on this organism in subsequent generations[31,41]. Host-associated microbiotas can establish modes of ecological inheritance in at least two ways.

First, by contributing to host trait variation, microbiotas can facilitate the inheritance of natural selection pressures on host genetic variation associated with that trait (Fig. 1, number (3)). Consider a host trait $z$ that is additively determined by a genetic component $g$ and a microbiota component $m$, such that $z = g + m$. This model of host trait architecture has previously been applied theoretically by ref. 34, and expanded on by ref. 35. Writing $W(z)$ for host fitness as a function of host trait, selection on the host trait is defined as the covariance of fitness and phenotype: $S_z = \mathrm{cov}(W, z)$. This covariance can be partitioned into selection on host trait components $g$ and $m$, given by $S_g = \mathrm{cov}(W, g)$ and $S_m = \mathrm{cov}(W, m)$, such that $S_z = S_g + S_m$ (Fig. 1; number (3)). This illustrates that selection at the level of host trait results in indirect selection at the levels of host genotype and host microbiota. Rearranging these components shows that selection on host genotype is mediated by selection on host microbiota (that is, $S_g = S_z - S_m$). Hence, in this case, the inheritance of microbiota variation across host generations establishes a mode of ecological inheritance.

Second, microbiotas can modify selection pressures on host variation by altering the host environment, such as resource availability and habitat quality (Fig. 1, number (4)). These modified selection pressures can have long-term evolutionary consequences on host traits, such as immunological profiles, tissue structures and physiological processes that influence specific microbial functions[18]. Summarizing the effect of the environmental microbiota on host fitness as $m_E$, we can include it as a parameter of the fitness function: $W(z \mid m_E)$. Then, the correlation of $m_E$ between host generations maintained by host activity (such as shedding) results in the maintenance of selection pressures on host trait variation, and therefore establishes a second mode of ecological inheritance. We note that the focal host trait does not need to be mediated by the host microbiota. Empirically, $m_E$ can be measured as the effect of the microbiota in the host's immediate environment on host fitness obtained from a linear model that explains the variance of host fitness, both in terms of host traits and in terms of the environmental microbiota composition.

## IGEs

IGEs are the influence of an individual's genotype on the phenotype of another (typically conspecific) individual[32]. As IGEs contribute to the expression and inheritance of phenotypic variation, they have important evolutionary consequences. A major application of IGEs is to understand the evolutionary consequences of social interactions in social insects[48–51].

Host-associated microbiotas establish IGEs between host individuals when three conditions are met: (1) host genes determine microbiota composition, (2) microbiota composition is transmissible between individuals, and (3) microbiotas mediate a host trait. When these conditions are met, microbiota transmission (which may occur during social encounters[52]) forms the mechanism for the genes of one host to influence the phenotype of another host (Fig. 2a).

To illustrate this, consider microbiota composition as a host trait $z_1$ that is mediated by host genotype $g_1$, such that $z_1 = g_1 + e_1$, where $e_1$ is an environmental effect. Now consider another host trait that is mediated by host genes and microbiota composition: $z_2 = g_2 + e_2 + z_1$. In this scenario, $z_2$ is a quantitative host character that is partially mediated by its microbiota such that $z_1$ and $g_2$ are obtained as the additive microbiota and genetic effects given by a linear model, with $e_2$ taken as a standard error term[40]. Furthermore, with measurements of $z_1$ across hosts in hand, $g_1$ and $e_1$ can then be obtained as the host additive genetic effect and error term, respectively, for a lineal model of $z_1$. This IGE model is similar to the starting point taken by ref. 53 to derive their model for interactions with non-reciprocal effects.

Assuming that each individual engages in a single interaction with another randomly chosen individual, we denote by $z_1', g_1', e_1'$ the microbiota trait and trait components for the non-focal interacting individual. Suppose that the social interaction results in an exchange of microorganisms, so that their microbiota composition traits become similar by $0 \le \psi \le 1$. Writing $z_1^*$ as the microbiota composition of the focal individual after the interaction, we then have $z_1^* = (1-\psi)z_1 + \psi z_1'$. Assuming $z_2$ is expressed after microorganism exchange, we can write $z_2 = g_2 + e_2 + (1-\psi)z_1 + \psi z_1'$. The coefficient $\psi$ measures microbial transmission via social contact, quantifying an indirect genetic effect of the interacting partner's additive genetic value $g_1'$ on the expression of the focal individual's trait value $z_2$.

The IGE framework has applications for understanding host–microbiome evolution, particularly for systems where social transmission of microorganisms (either through direct contact or through spatial proximity) between unrelated individuals has an important role. For instance, because conspecific social interactions are known to structure gut microbiotas in baboons[54] and wood mice[55], the IGE framework may provide important insights into the roles that host genetics and microbiotas have in mediating host traits for these systems. As the IGE framework considers interactions between arbitrary individuals and only incorporates non-random interactions mediated by trait covariances, and not necessarily relatedness[32], it requires additional assumptions to apply to systems with substantial parent–offspring microorganism transmission. To model this complementary scenario more directly, the related framework of MEs has greater utility.

## MEs

An ME is the influence of a parental phenotype on an offspring phenotype, controlling for genetic variation, mediated by parent–offspring interactions such as maternal care[56]. Host-associated microbiotas establish MEs between host parents and host offspring when a host trait is mediated by its microbiota, and part of the host's microbiota is inherited from direct or indirect parent–offspring transmission (Box 1 and Fig. 2b). For instance, the composition of mammalian milk is mediated by host microbiotas and is transmitted directly from parent to offspring[57].

Assume a host trait decomposes as $z = g + m$, where $g$ is the additive host genetic effect and $m$ is the additive effect of host microbiota composition. Just as with the niche construction framework, $m$ can be measured as the additive effect of microbiota composition on host trait obtained from a linear model[35,40]. To account for microbiota transmission directly from parents to offspring, we model the offspring microbiota $m'$ as depicted in Fig. 2b, where $\iota$ is the proportion of the offspring microbiota inherited from its parent, and $(1-\iota)$ is the proportion acquired from the environment and unrelated hosts, together having the average composition $\xi$. $\delta$ is an ontogenetic differential of the microbiota that is independent of $m$.

With this, we can model microbiota-mediated MEs for a wide range of maternal care patterns (Box 1 for examples). The microbiota's contribution to an ME is measured as the partial regression coefficient of offspring trait on maternal trait, holding genetic variation constant[30]. Then, writing the trait variance as $\mathrm{var}(z)$, and $\mathrm{var}(m)$ as the component of $\mathrm{var}(z)$ explained by microbiota variation, the ME is then $\omega = \iota\,(\mathrm{var}(m)/\mathrm{var}(z))$, which is the slope obtained from regressing the offspring trait $z'$ on the parental trait $z$ after controlling for genetic variation (Fig. 2b).

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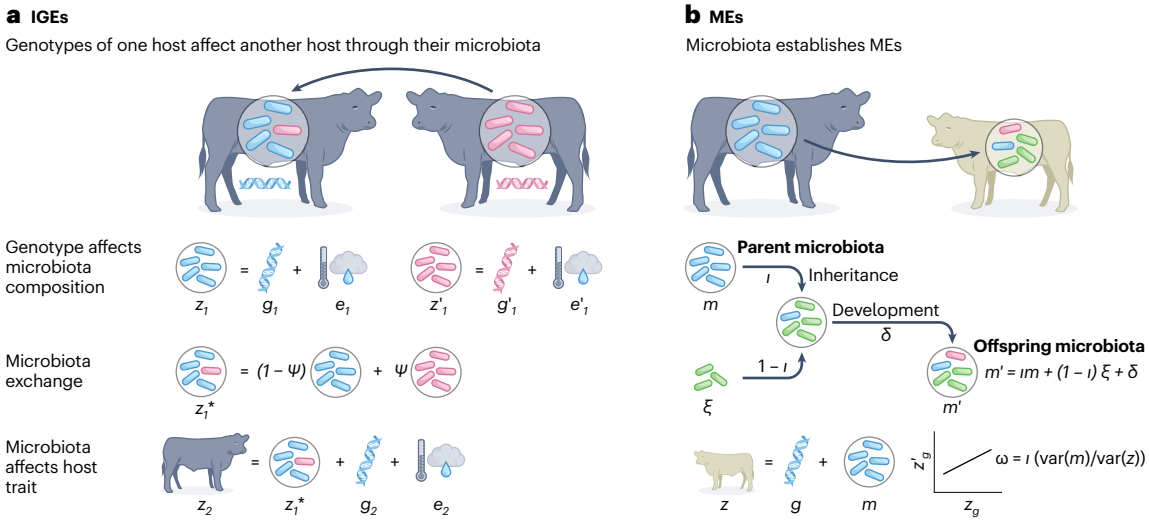



**Fig. 2 | A model of microbiota-mediated IGEs and MEs. a**, Microbiota composition of a focal host and interacting partner ($z_1$ and $z'_1$, respectively) are mediated by host genetic effects ($g_1$, $g'_1$) and environmental effects ($e_1$, $e'_1$). Updated compositions after microorganisms are exchanged via social contact, quantified by $\psi$, are denoted by $z_1^*$. Then a second trait $z_2$, mediated by microbiota composition $z_1^*$ (in addition to genetic effect $g_2$ and environmental effect $e_2$), is affected by the genetic factor $g'_1$ in the interacting partner. **b**, At birth the host offspring acquires a proportion $\iota$ of its microbiota from the parent, and the remaining proportion $1-\iota$ from the environment (with composition $\xi$). The offspring microbiota subsequently shifts by an amount $\delta$ throughout development. Quantification of the microbiota-mediated MEs is illustrated in the bottom section as the slope $\omega$ of the relationship between offspring trait value $z'_g$ and parental trait value $z_g$, where the subscript $_g$ indicates host genetic variation is held constant.

This shows that host-associated microbiotas can be modelled as maternal characters. The application is limited to microbiome inheritance resulting from strict parent–offspring transmission. Furthermore, MEs on microbiota composition will probably depend on organism age and diminish as offspring acquire microorganisms from their environment and unrelated conspecifics[58] and as within-host microbiota dynamics unfold[25]. By contrast, estimates of MEs on some microbially mediated traits may actually become stronger with age as the early microbiota has an important role in development[59].

## MLS

This last framework applies to host–microbiota systems in which either microbiota-mediated host traits or microbial traits are heritable and subject to natural selection. Heritable variation in the host and microbiota fidelity across generations are essential for the host–microbe system to collectively respond to natural selection (Box 1). In other words, Lewontin's conditions[60] must be met. Given these restrictive conditions, most host–microbiota systems are not considered evolutionary individuals.

Host–microbiota systems show natural selection at the levels of individuals, kin and as groups of unrelated organisms[61], all explicitly captured by MLS models. As these models do not make a priori assumptions about inheritance, they are useful for understanding when a given inheritance mechanism establishes selection at higher levels of organization. MLS models have been categorized into two types: MLS1 explains differences among groups in their respective rates of production of contained individuals, while MLS2 explains differences among groups in their respective rates of production of distinct new groups[62,63].

By testing different MLS models, we can ascertain whether individual, kin, or group selection, or a combination, is the predominant force in the evolution of emergent host–microbe phenotypes. For instance, Hermsen[64] recently introduced a framework for measuring MLS1 and MLS2 to analyse simulated data. Next, we describe how this general framework may be operationalized for the measurement of MLS1 and MLS2 in the context of host–microbe systems.

So far, MLS in host–microbiota systems has not yet been empirically measured. However, MLS may be especially relevant in systems where hosts control the composition of functionally important internal microbiotas. For instance, in mammals, intestinal microbiota composition is shaped by within-host selection pressures due to the host immune system[65] or the epithelial environment[66]. These within-host filters generate microbiota variation among hosts, creating potential for selection to operate at both within-host and among-host levels. In Box 2, we illustrate a design for an experimental test of MLS1.

### MLS2

Due to its relative simplicity, we begin with MLS2 (Fig. 3), which focuses on the reproduction of groups. MLS2 occurs when host birth and death rates depend on microbiota composition. Focusing on a microbiota-mediated host trait $z$, the change in mean traits between parental and offspring generations can be expressed using the Price equation as

$$\Delta\bar{z} = S + T, \quad (1)$$

where $\Delta\bar{z} = \bar{z}' - \bar{z}$ is the difference between offspring mean trait $\bar{z}'$ and parental mean trait $\bar{z}$; host-level selection is captured by $S = \mathrm{cov}(W, z)/\overline{W}$, with $W$ ($\overline{W}$) host individual fitness (host mean fitness) and the covariance is taken across host parents. The term $T = \overline{W\Delta z}/\overline{W}$ is called transmission bias, and here accounts for trait differences resulting from within-host microbiota dynamics and differential colonization abilities of microorganisms. More precisely, $\Delta z$ is the average ontogenetic differential across offspring of a given host parent so that $\overline{W\Delta z}$ is the average ontogenetic differential across all offspring[67]. Following Hermsen's[64] interpretation, in the context of host–microbiota systems, $T$ captures within-host selection on microbial variation. As one could measure microbiota-mediated host traits and host fitness, at least in principle, this approach is particularly amenable to empirical study. In particular, for cases in which the mean trait differential $\Delta\bar{z}$ and host-level selection $S$ are directly measurable, the transmission bias term can be inferred as $T = \Delta\bar{z} - S$. This provides an explicit method to partition host trait dynamics into components due to selection among hosts and selection within hosts.

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## BOX 2

# Testing theory using the *Daphnia* host system

The water flea *Daphnia* (see the figure below), a freshwater crustacean, is a model organism in many biological fields, such as ecotoxicology[118], epidemiology[119] and eco-evolutionary dynamics[120]. Increasingly, it is also used to study host–microbiota evolution[121]. Several features make the *Daphnia*–microbiota system uniquely suited to test theoretical predictions:

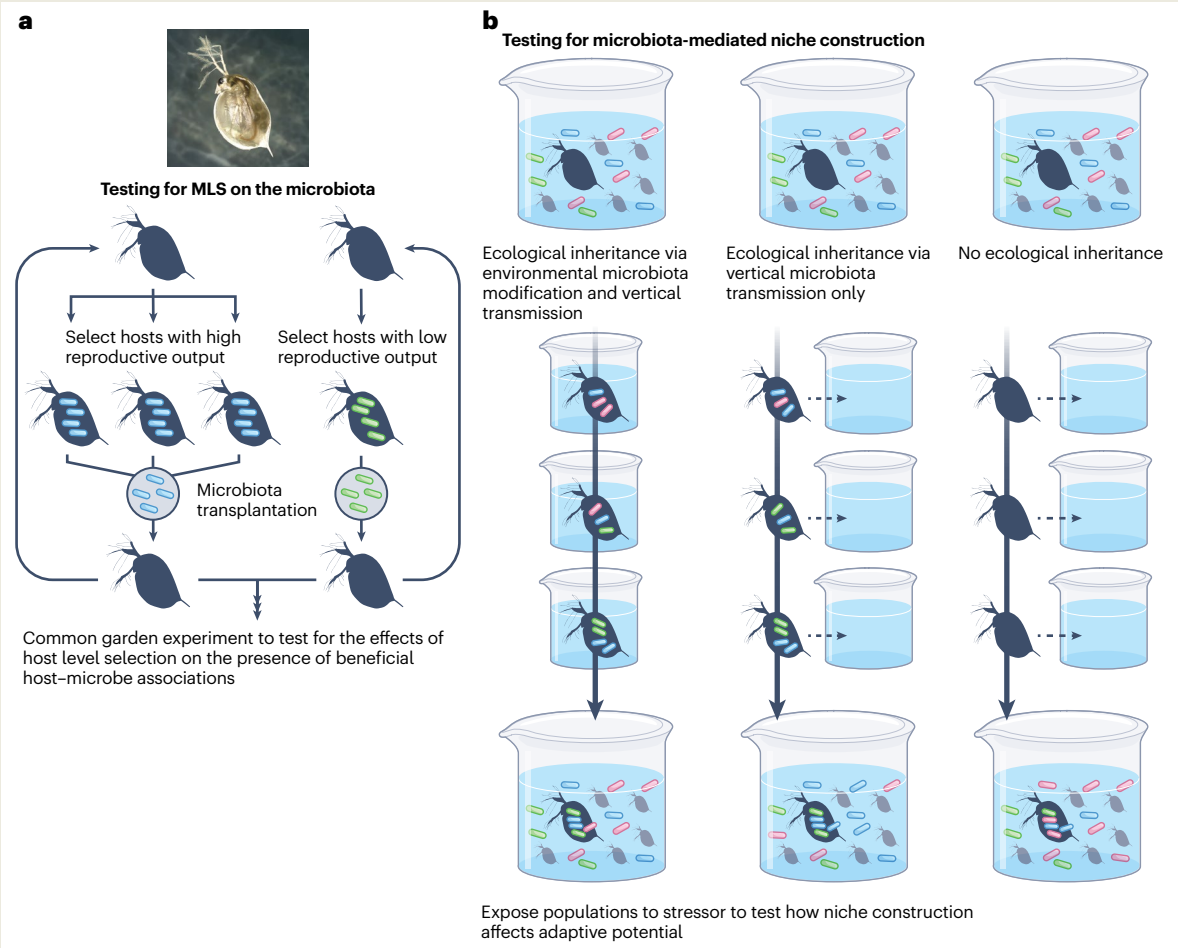
- Its gut microbiota is relatively simple, has distinct core members and is clearly separated from the surrounding aquatic microbiota[122].
- Microbiota composition is shaped by both environmental conditions (for example, temperature[123]) and host genotype[124].
- The microbiota influences host fitness[125,126] and may mediate plastic responses to the environment[127].
- Many phenotypic traits—such as defence induction[128]—are well characterized and may be partly microbiota mediated.
- Both horizontal and vertical microbiota transmission occur[129,130].
- Experimental manipulations (for example, gnotobiotic rearing and microbiota transplants) are feasible, and host genotypes can be controlled via clonal lineages[126,131].

These advantages enable rigorous tests of theoretical frameworks. We next outline two illustrative experimental designs:

### Testing niche construction

*Daphnia*–microbiota associations can be evolved in mesocosms under three treatments (right-hand side of the figure below):

1. Environmental microbiota inheritance: hosts and their microbial environment are maintained together, allowing microbial shaping of the environment and ecological inheritance via environmental modification.




**Experimental approaches for testing MLS1 and evolutionary niche construction using the water flea *Daphnia* system. a**, MLS1 can be tested by maintaining two treatments: microorganisms from the hosts with highest fitness are isolated and transplanted into the subsequent host generation (left) and microorganisms from hosts with the lowest fitness are isolated and transplanted into the subsequent host generation (right). **b**, Evolutionary niche construction can be tested by maintaining three treatments: mesocosms are maintained including host-associated and environmental microorganisms (left), host-associated microorganisms are included but environmental microorganisms are excluded (middle), and both host-associated and environmental microorganisms are excluded (right). Subsequently, host populations are exposed to a stressor to test whether ecological inheritance via environmental modification (left) or vertical transmission (middle) promote host adaptation compared with the absence of ecological inheritance (right). Photograph credit: Héléne Vanvelk and Maxime Fajgenblat (KU Leuven).

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2. Vertical microbiota transmission only: hosts are transferred to fresh environments regularly, preserving vertical transmission but preventing environmental shaping.
3. No ecological inheritance: both host-associated and environmental microorganisms are excluded at each transfer; microbiotas are provided via inoculum from naive *Daphnia*.

Each treatment is subjected to an environmental stressor (for example, toxins or elevated temperature) to:

- Evaluate whether niche construction promotes adaptation (via host reproduction or population growth)
- Investigate ecological and molecular changes (for example, microbiota dynamics, or host or metatranscriptomic responses)

A follow-up transplant experiment can assess whether evolved hosts benefit specifically from shaped microbial environments.

**Testing MLS1**

To test MLS1, microbiotas can be evolved under two selection regimes (left-hand side of the figure below):

1. Among-host selection: microorganisms are transferred from the most reproductively successful hosts, favouring mutualistic strains that benefit both host and microorganism.
2. Within-host selection: microorganisms are transferred from hosts with low reproductive output, favouring strains that enhance their own replication at a potential cost to the host.

After experimental evolution, microbial communities are introduced to a standardized host population. Differences in host fitness and microbial load can reveal the relative strength and outcome of different levels of selection.

### MLS1

MLS1, in this context, takes a microorganism-centric approach and occurs when microorganism strains or species show differential fitness[64]. It considers traits of different microorganism taxa (for example, strains or species), which we write as $\alpha_i$ for microorganism taxon $i$. These traits may be any phenotype of the microorganism, such as growth rates on different carbon sources or the additive effect on a host trait. Selection at different levels is then quantified using covariances that involve the microbial trait and a notion of microorganism fitness ($w_i$ for microorganism $i$). Here we focus on two measures of microorganism fitness: host colonization rates and within-host proliferation rates.

First, we can quantify microorganism colonization fitness for a given host by comparing the relative abundances of microorganisms in the immediate environment of the host (for example, the substrate and other hosts) to the relative abundances of microorganisms inhabiting the host (Fig. 4; top). The relative abundance of microorganism $i$ in the host can be written as $\kappa_i' = w_i\kappa_i$, with $\kappa_i$ being the relative abundance of microorganism $i$ within the immediate environment of the host. As $\kappa_i$ and $\kappa_i'$ are, at least in principle, measurable, we can quantify microorganism colonization fitness as $w_i^{(\kappa)} = \kappa_i/\kappa_i'$, where $(\kappa)$ emphasizes that colonization is used as a fitness measure.

Second, we can quantify microorganism growth fitness within a given host by comparing the relative abundances of microorganisms at host adolescence to relative abundances at host maturity. We write $\gamma_i$ for the relative abundance of microorganism $i$ in an adolescent host, and $\gamma_i'$ for its relative abundance when that same host has matured (Fig. 4; bottom). Growth fitness of microorganism $i$ can be quantified as $w_i^{(\gamma)} = \gamma_i/\gamma_i'$.

Regardless of the exact fitness measure, MLS1 suggests that selection on microorganism trait $\alpha$ (across the entire host–microbiota system) can be partitioned into two components: within- and among-host selection. Within-host selection ($S_{\text{within}}$) is found by first calculating the covariance of fitness and phenotypes across microorganisms within each host $j$ separately ($\text{cov}_j(w,\alpha)$), and then this covariance is averaged across hosts so that

$$S_{\text{within}} = \overline{\text{cov}(w,\alpha)}. \quad (2)$$

Among-host selection ($S_{\text{among}}$) is found by first averaging fitness and microorganism traits across microorganism taxa within each host separately (which we write as $\bar{w}$ and $\bar{\alpha}$, respectively, for a given host) before computing their covariance across hosts such that

$$S_{\text{among}} = \text{cov}(\bar{w},\bar{\alpha}). \quad (3)$$

This approach allows us to connect levels of selection occurring simultaneously across distinct scales of biological organization. More than two levels of organization can be considered by generalizing Hermsen's[64] framework, allowing for structured populations of host–microbe groups and within-host microorganism groups, which is crucial for the application of MLS to real-world systems.

## Discussion

### Limitations of existing frameworks

Each of the discussed pre-existing frameworks is useful for understanding specific cases of host–microbiome evolution, and one useful strategy is to select the framework that fits best to the study question and model system used. At the same time, some fundamental properties of host–microbiota systems necessitate expanding these frameworks. Further, these frameworks are not mutually exclusive and, in real-world scenarios where multiple processes and mechanisms act simultaneously, different frameworks will probably interact. For example, niche construction has clear applications for understanding the relationship between host and environmental microbiotas, and consequential selection of host–microbiota systems. Modification of the social environment via microbiota transmission may be considered a form of niche construction but this framework does not focus on microbiota transmission. The frameworks of IGEs and MEs are particularly useful for understanding the evolutionary consequences of such social and parent–offspring transmission, respectively. In systems with mixed modes of microbiota transmission, both IGEs and MEs jointly affect the dynamics and could be combined to account for such mixed transmission. Still, IGEs and MEs treat the consequences of microbiota transmission as fixed effects, limiting their ability to incorporate microbiota community dynamics, host immune response and variation of transmission other than what is explained by trait covariances.

MLS is useful as an overarching framework for understanding selection on complex host–microbiota systems. However, it is still under debate whether MLS1 and MLS2 are inequivalent, as the Price equation has supported their equivalency since the 1970s[63,68–70]. Further, although some group-level phenotypes may have relevance at the individual level, not all will. For example, metabolic complementation between deep sea clams and their chemosynthetic gill endosymbionts for haemoglobin-transported sulfur-based carbon fixation is selected for in the host–microbe assemblage, but not in the individual organisms, because the individual bivalve and bacterial genomes lack genes to complete the pathway[71].

### Opportunities for developing new frameworks

Host–microbiota systems provide a number of exciting opportunities for extending and developing theoretical approaches to describe features that are not sufficiently captured by our focal frameworks, as is also apparent from the previously described limitations of these frameworks.

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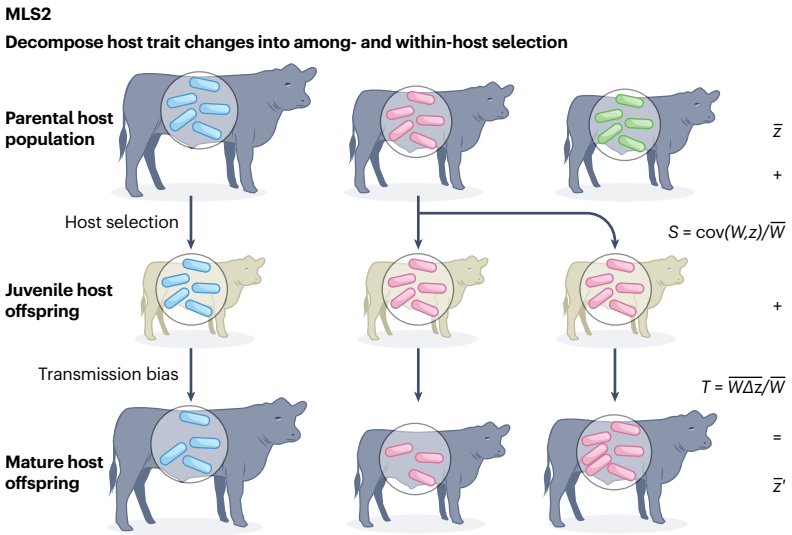



**Fig. 3 | MLS2 in the context of host–microbiota systems.** The top row represents the parent population with mean trait $\bar{z}$. Selection among hosts occurs via differential reproduction, forming host offspring in the middle row. Host-level selection is quantified by $S = \mathrm{cov}(W,z)/\bar{W}$, where $W$ is host fitness and $\bar{W}$ is mean fitness in the parental population. Subsequently, while offspring develop, transmission bias occurs due to within-host microbiota dynamics, and this leads to the mature offspring population (bottom row) with mean trait $\bar{z}'$. Transmission bias, capturing within-host selection, is quantified by $T = \overline{W\Delta z}/\bar{W}$, where $\Delta z$ is the average ontogenetic differential (due to within-host microbiota dynamics) across offspring of a host parent and $\overline{W\Delta z}$ is the average of the product $W\Delta z$ across host parents. Note that transmission bias is depicted as differential microorganism abundance, but it could also arise from compositional divergence.

For example, microbiota composition varies over the course of a host's life[72,73]. Here, theory on ontogenetic changes in maternal and genetic contributions to host phenotypic variation[74] may provide useful insights to microbiota changes during host development and its implications for responses to selection. For instance, a maternal signal in microbiota composition that diminishes with host age[25,75] could be captured by a negative relationship between host age and the contribution of MEs. Further, host microbiota composition is shaped by fluctuating microbial abundances resulting in within-host ecological interactions, but such interactions are ignored in the NGI and MLS frameworks that we discussed. Further, microbiota vary across host organs[75–77] but the frameworks we present do not account for this within-host spatial variation.

These frameworks can be extended to account for such ecological details by integrating models of microbial community dynamics into host trait architecture. These biotic interactions are even further complicated by the existence of multiple trophic levels within a microbiota community (for example, interactions of bacteria with phages or predatory bacteria)[78–80]. Microorganisms can also show context dependence in their contributions to host fitness, where they act as mutualists in one environment but as pathogens in another[81–84]. Theory on fluctuating selection[85] could be used to assess host–microbiome evolution in such a case.

In addition, there are many opportunities to expand the theory of MLS. For example, stochastic simulation of MLS processes could enable the development of new MLS models and theories to pinpoint the conditions required for cooperation among microorganisms and hosts to evolve[86]. Spatial structure shapes the formation of groups that can respond to selection pressures[64,69,87,88], emphasizing the need to incorporate environmental parameters in MLS models. Genetic models for mapping trait selection onto the complex genetic basis for that trait[89] could be used to map group-level phenotypic selection onto individual genotypes. Incorporating genetic parameters into MLS models will enable the use of genome-wide datasets. Further, the effect of host versus symbiont population size and generation time on the rates of coevolution should be considered[26,90].

Lastly, it may be useful to consider microbiota-mediated host traits as a form of phenotypic plasticity, with host microbiota composition as the environmental factor. However, as microbiota are transmissible between host individuals within a generation, and because theoretical approaches to study reaction norm evolution have not focused on transmissible factors[91–95], there is a need to extend reaction norm theory to account for the complexity of host–microbiota systems. Microbiota transmission, either directly from parent to offspring as in the case of maternal characters or socially as in the case of IGE, can explain patterns of microbiota variation that then feed back on host–microbiome evolution[52]. Moreover, microbiota-mediated plasticity can act at different levels and timescales[96]. For example, a new environmental challenge can be accommodated fastest by ecological changes in microbiota community composition, followed by evolutionary genetic changes in single microbial lineages. Such microbiota plasticity can further selectively favour hosts that either select the beneficial microorganisms from the environment or ensure their vertical transmission, as a kind of microbiota-mediated Baldwin effect[96]. Hence, application of the phenotypic plasticity framework to host–microbiome evolution would need to be extended to account for environmental factors that are transmissible between hosts within and across generations. Doing so will be useful for understanding microbiota-mediated adaptation to new environments[11] and microbiota-mediated local adaptation[10].

## The need for 'empirically friendly' theory

We feel that it is important to recognize that existing theoretical frameworks were often developed with biological systems in mind other than host–microbiota systems[97]. Not only has this resulted in frameworks that lack key aspects of the biology of host–microbiota systems (as we discuss in the previous section) but it has also limited the application of these frameworks to host–microbiota systems in the laboratory and the field. To make these frameworks maximally useful, it is important that the validity of the underlying assumptions of these frameworks is determined empirically. It is also crucial that these frameworks are constructed in a way that makes their predictions empirically testable, given the technical limitations of empirical microbiota research. For example, for many host–microbiota systems, empiricists are limited to surveying relative abundance or presence versus absence of microbial taxa, and for theory to be maximally useful it must generate predictions

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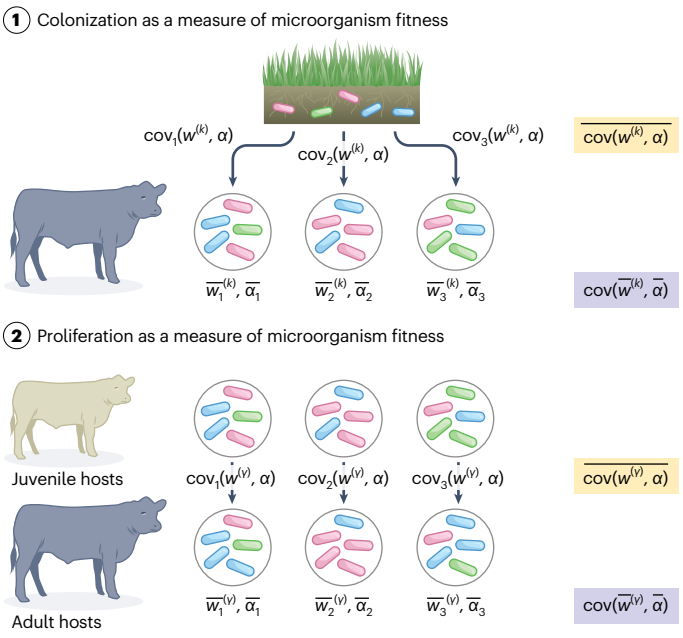



**Fig. 4 | MLS1 in the context of host–microbiota systems.** Circles represent host individuals. Microorganism individuals are represented by capsule shapes, and their trait values $\alpha$ are indicated by colour (red, blue or green). Top: number (1) illustrates how selection at the levels of host and microorganisms can be measured when microbial fitness $w$ is taken as the colonization fitness $w^{(\kappa)}$ (see main text). Here microorganisms in the immediate host environment (depicted by the grass and soil at the top of the figure) differentially colonize hosts (represented by arrows following microorganisms from the environment into host individuals). Microorganism-level selection resulting from these differential colonization abilities for the $j$th host is quantified by $\mathrm{cov}_j\left(w^{(\kappa)}, \alpha\right)$, and these selection indices are then averaged across hosts to obtain a global measure of microorganism-level selection $\overline{\mathrm{cov}}(w^{(\kappa)}, \alpha)$ (highlighted in beige). Host-level selection is quantified by first averaging microorganism fitness and trait values within each host (respectively, $\bar{w}_i$ and $\bar{\alpha}_i$ in host $i$) before taking their covariance across hosts $\mathrm{cov}(\bar{w}^{(\kappa)}, \bar{\alpha})$ (highlighted in purple). Bottom: number (2) illustrates how these levels of selection may be quantified when using proliferation as a measure of microbial fitness. Juvenile hosts are represented by the top row, and vertical arrows connect them to their matured selves. Here, changes in host microbiotas are presumed to be a consequence of within-host microbiota dynamics, quantified by the growth fitness $w^{(\gamma)}$ (see main text).

for these microbiota attributes. Tailoring theory in this way will probably require direct collaborations between theoretical and empirical microbiota scientists to enable an iterative refinement of theory with information from actual host–microbiota systems. This requires host–microbiota systems that are sufficiently tractable to measure relevant quantities, such as correlations among microbiota composition, host traits and host fitness. Example systems include the water flea *Daphnia* (Box 2), the nematode *Caenorhabditis elegans*[13,98], the zebrafish *Danio rerio*[99], the bobtail squid *Euprymna scolopes*[100] or insect–*Wolbachia* associations[101].

## Conclusion

We presented four frameworks developed in the fields of evolutionary biology that help to generate new insights into host–microbiome evolution. To capture the biological diversity of such host–microbe systems and produce empirically testable predictions, these frameworks require thoughtful expansion and, in some cases, the development of new theory, in close collaboration with empirical microbiota scientists. We envision that the initial result will be a mosaic of theoretical frameworks, each tuned to the set of processes considered and questions asked, with the initial goal of clarifying concepts. Such a mosaic could eventually lead to the identification of general principles underlying the interactions between microorganisms and their animal and plant hosts, greatly expanding our understanding of the evolutionary consequences of the host–microbe associations omnipresent across the tree of life.

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## Author contributions

B.W., S.L.R., H.S., B.J.M.B. and M.B. conceived the ideas. B.W. developed the methodology. M.B. created the visualizations. B.W., S.L.R. and M.B. wrote the first draft of the paper. All authors contributed substantially to revisions.

## Competing interests

The authors declare no competing interests.

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